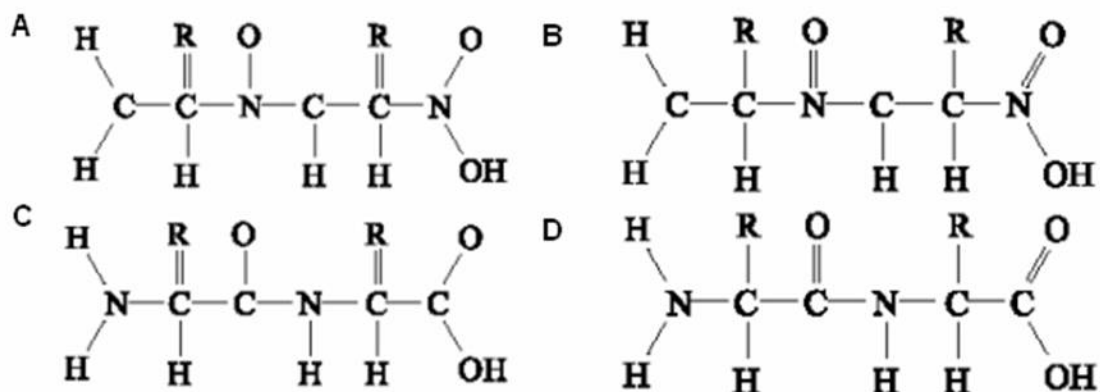


1. From which cell organelle are nucleic acids absent?

- A Chloroplast
- B Mitochondrion
- C Ribosome
- D Smooth endoplasmic reticulum

2. Which is the structure of a dipeptide? D



3. Three pure substances are analysed and the elements that each contains are listed in the table below:

	C	H	O	N	S	P
E	+	+	+	-	-	-
F	+	+	+	+	-	-
G	+	+	+	+	+	-

Which combination could be correct?

- | | | | |
|----------|------------|--------------|--------------|
| | E | F | G |
| A | adenine | carbohydrate | fat |
| B | fat | adenine | amino acid |
| C | amino acid | fat | carbohydrate |
| D | fat | carbohydrate | amino acid |

4. Colchicine is a chemical substance that prevents the formation of microtubules. Which phase of mitosis would be prevented if the dividing cells were treated with colchicine.

- A Disintegration of the nuclear membrane.
- B Replication of DNA.
- C Separation of genetically identical chromosomes (chromatids).
- D Supercoiling of chromosomes.

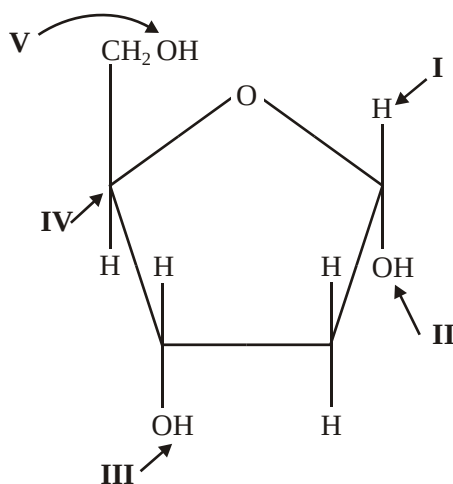
5. The following events occur during the mitotic cell cycle.

- 1 Division of centromeres
- 2 Duplication of centrioles
- 3 Duplication of chromosomes
- 4 Overlapping of non-kinetochore spindle fibres

Which two events ensure that the daughter cells are genetically identical to each other and to the parent cell?

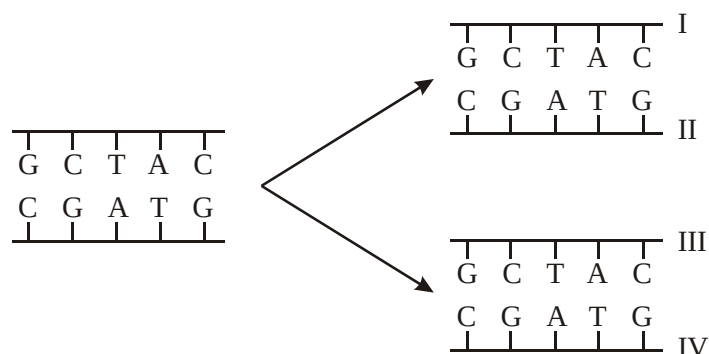
- A 1 and 2
B 1 and 3
C 2 and 3
D 3 and 4

6. To which parts of the deoxyribose molecule do phosphates bind in DNA?



- A I and V
B III and IV
C II and III
D III and V

7. The diagram below shows a short section of DNA molecule before and after replication. If the nucleotides used to replicate the DNA were radioactive, which strands in the replicated molecules would be radioactive?



- A** II and III only
B I and III only
C I and II only
D I, II, III and IV

8. The steps involved in DNA replication and protein production are summarised below.

Step A: DNA is copied and each new cell gets a full copy.	] DNA replication
Step B: Information is copied from DNA and taken to cytoplasm.	
Step C: Ribosome reads information and assembles protein.	] Protein synthesis
Step D: Protein formation is completed.	

In which step would a mutation lead to the formation of a new allele?

- A** Step A
B Step B
C Step C
D Step D
9. An mRNA codon for the amino acid alanine is GCC.
 How many alanine molecules are present in the polypeptide, containing eight amino acids, coded for by the following DNA template?

DNA template: TCGGCCTACCGGGCCCATGCCAAT

- A** 0
B 2
C 1
D 3

10. The following events occur in the replication of DNA:

1. bonds between complementary bases break
2. bonds between complementary bases form
3. DNA molecule uncoils
4. Opposite strands separate
5. Sugar-phosphate bonds form
6. Free nucleotides align with complementary nucleotides on each strand

In which order do these events take place?

A	1	3	6	4	2	5
B	3	1	4	6	2	5
C	3	6	1	4	5	2
D	4	3	1	6	5	2

11. Amino acids complex with transfer RNA through

- A an aminoacyl link with cytosine residue at the 5' end of the chain
- B a phosphodiester bond
- C hydrogen bond
- D an aminoacyl link with a 3' terminal adenosine residue

12. An E. coli cell is grown in the presence of high amounts of glucose. Which of the following is true?

- A The cell will utilize lactose as a carbon source exclusively
- B The level of cyclic-AMP in the cell will be low
- C The level of cyclic-AMP in the cell will be high
- D Transcription of mRNA from the lac operon will be high

13. In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly(A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP). Luciferase activity would therefore be a reflection of the amount of proteins produced (translation efficiency).

Table 3.1 below illustrates the effects of capping and polyadenylation on the life-span of the luciferase mRNA and its protein products.

Table 3.1

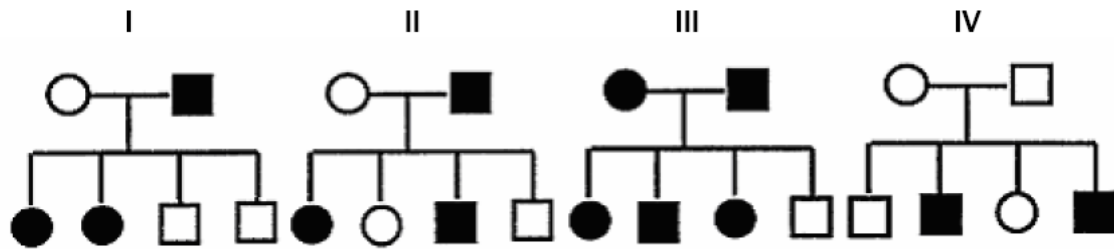
mRNA	Luciferase mRNA half life (min)	Luciferase activity (units)
<u>Uncapped</u>		
Poly(A) tail absent	31	2941
Poly(A) tail present	44	4480
<u>Capped</u>		
Poly(A) tail absent	53	62595
Poly(A) tail present	100	1331917

Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount.

Which of the following is **true**?

- A** Capping and polyadenylation have the same effect on translational efficiency.
 - B** In capped mRNA, presence of poly(A) results in an increase in luciferase activity of 10 times.
 - C** Polyadenylation protects against mRNA degradation in mRNA more than capping.
 - D** In polyadenylated mRNA, capping the mRNA results in an increase in luciferase activity of 297 times.
14. How many different genotypes can be found in the offspring of the cross **BbCc** x **BbCc**?
- A** 2
 - B** 4
 - C** 9
 - D** 16

15. The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male. Which pedigree(s) represent(s) an inheritance of dominant sex-linked trait?



- A I only
B I and III
 C II and III
 D III and IV

16. In the Sailfin Molly fish, gold colour (g) is recessive to normal colour (G). When a gold coloured fish was crossed with a normal colour fish, 55 of the offspring were normal colour and 45 were gold colour.

One hypothesis is that the normal parent was heterozygous.

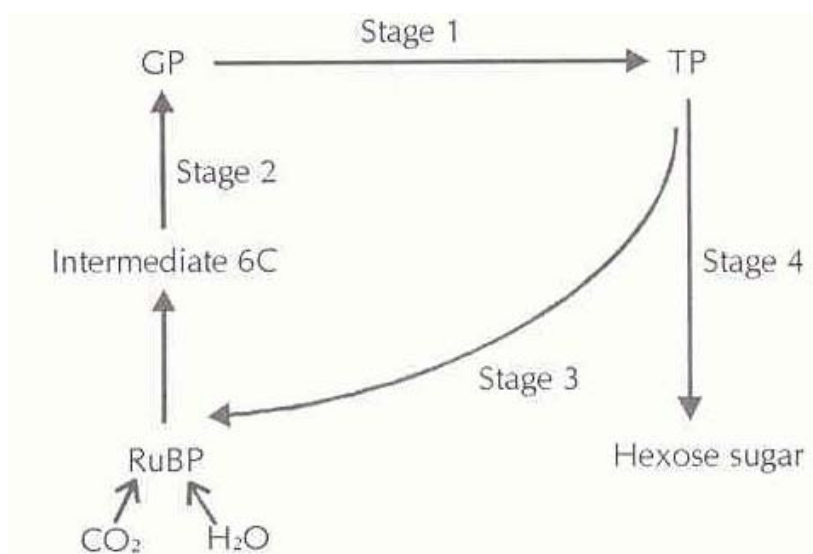
The Chi-squared value is 1. The table below shows the probability values.

Degrees of freedom	Probability			
	0.99	0.950	0.05	0.01
1	0.000	0.004	3.84	6.64
2	0.020	0.103	5.99	9.21
3	0.115	0.352	7.82	11.35

Which of the following is the correct response?

- A** Do not reject the hypothesis because the Chi-squared value is less than 3.84.
 B Reject the hypothesis because the probability is less than 0.05.
 C Do not reject the hypothesis because the Chi-squared value is less than 5.99.
 D Reject the hypothesis because there is not enough evidence.

17. The diagram below shows the stages in the Calvin cycle. Which of the stages use ATP?



- A** Stage 1 and Stage 3
- B Stage 2 and Stage 4
- C Stage 1 and Stage 2
- D Stage 3 and Stage 4

18. Which of the following occurs in the light reaction of photosynthesis?

- A** ADP is phosphorylated
- B Hexose phosphates are hydrolysed
- C Reduced NADP is oxidised
- D Triose phosphates are reduced

19. If light intensity is uniformly high, under which of the following conditions is carbon dioxide most likely to be the limiting factor upon the rate of photosynthesis?

	Temperature, °C	% Carbon Dioxide
A	25	4.00
B	15	0.04
C	5	0.04
D	5	0.01

20. What is the role of oxygen in cell respiration?

- A** to act as the final hydrogen acceptor in the electron transport chain
- B to combine with acetyl coenzyme A to form citrate in the Krebs cycle
- C to combine with carbon from carbohydrates to form carbon dioxide
- D to oxidise reduced NAD in oxidative phosphorylation

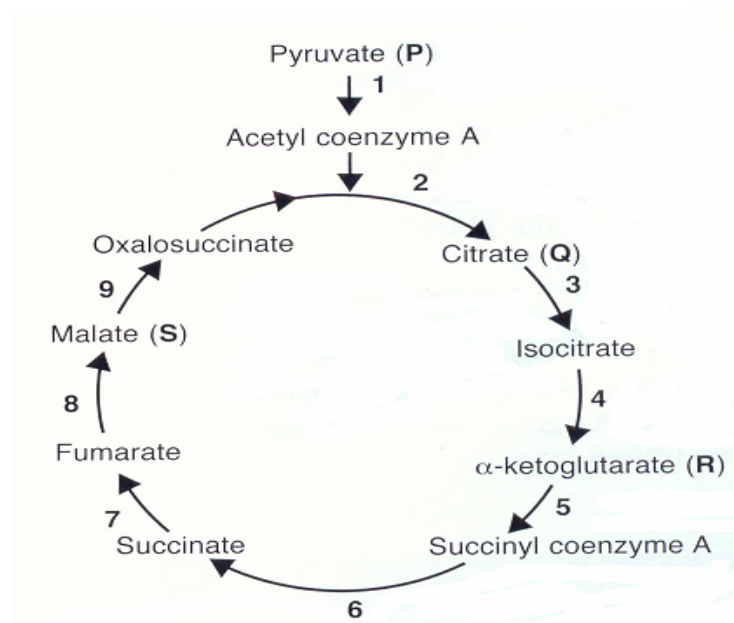
21. According to the chemiosmotic hypothesis,

- A H^+ ions diffuse out of the mitochondrion; this movement releases energy that can be used to make ATP.
- B H^+ ions are actively transported from the cytoplasm into the inner compartment of the mitochondrion.
- C ATP is produced in the mitochondrion when the hydrogen ions stored in the outer compartment flow back into the inner compartment.
- D the electron-transport molecules synthesize ATP.

22. What products are formed when anaerobic respiration takes place in muscles and yeast cells.

	Muscles	Yeast Cells
A	Pyruvate, NAD^+ , ATP	Ethanol, NAD^+ , ATP
B	Ethanol, NADH, ADP, CO_2	Lactate, NADH, ADP, CO_2
C	Lactate, NAD^+ , ATP, CO_2	Ethanol, NAD^+ , ATP
D	Lactate, NAD^+ , ATP	Ethanol, NAD^+ , ATP, CO_2

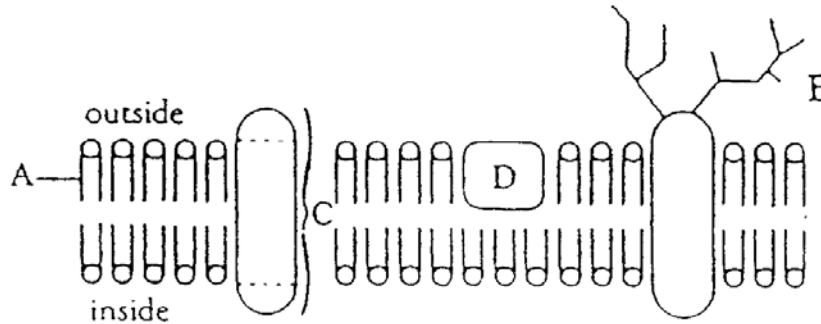
23.



At which stages are hydrogen atoms transferred to NAD^+ ?

- A 1, 4, 5, and 7
- B 1, 4, 5, and 9
- C 2, 4, 5, and 7
- D 4, 5, 7, and 9

24. Which structure(s) contain(s) the presence of a peptide bond?



- A D only
- B A and B
- C C and D**
- D B, C and D

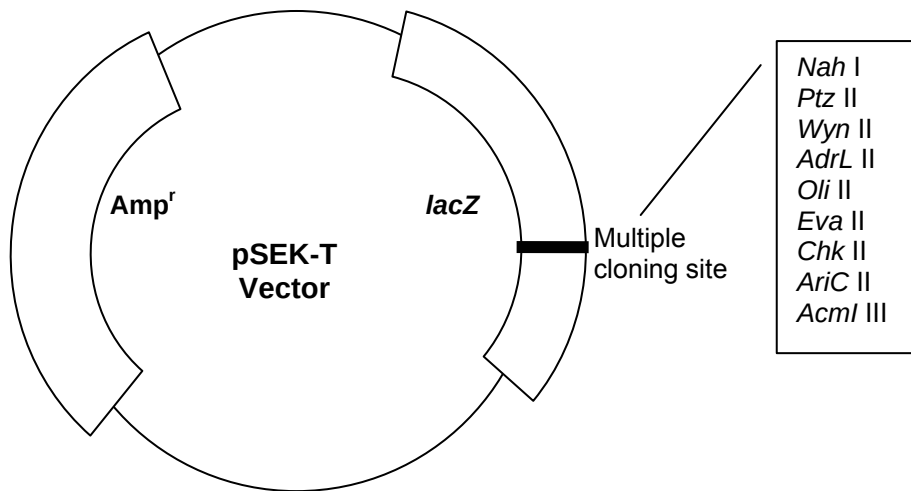
25. Why has evolution resulted in the appearance of antibiotic resistant bacteria?

- A Bacteria learn the ability to neutralize the effects of antibiotic and they pass this trait to the next generation.
- B Bacteria modify their metabolism to cope with the presence of antibiotics.
- C Bacteria resistant to the antibiotic survive and pass on this characteristic to their next generation.**
- D Bacteria develop resistant due to the incomplete course of antibiotic.

26. Which of the following situation would evolution be fastest for an interbreeding population?

	Migration	Selection pressure	Variation due to mutation
A	High rate	High	High
B	High rate	Low	High
C	Low rate	High	High
D	Low rate	Low	Low

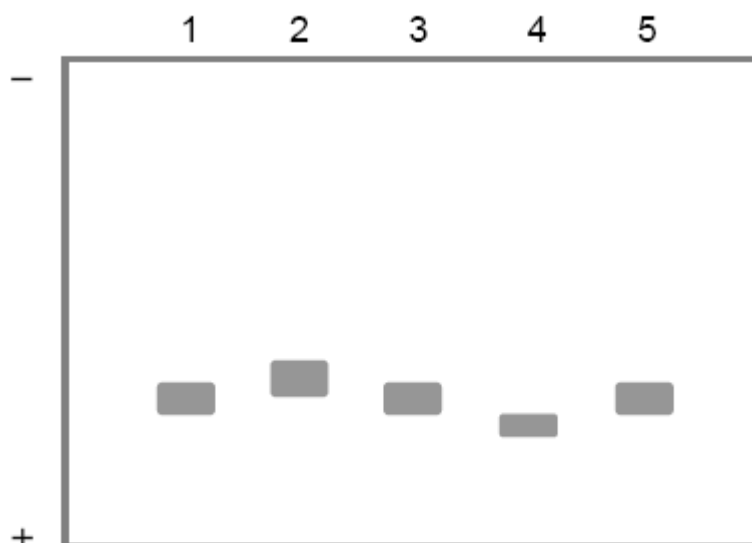
27. As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the D-ONG protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for D-ONG protein were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	Chemicals to be added		Colour of colonies
A	Ampicillin	X-gal	Blue
B	β -galactosidase	X-gal	Blue
C	Ampicillin	X-gal	White
D	β -galactosidase	lactose	White

28. The diagram shows the results of SDS polyacrylamide gel electrophoresis of the same protein extracted from five different individuals.



Which of the following correctly explains why the proteins extracted from individuals 2 and 4 did not travel the same distance on the gel as those extracted from individuals 1, 3 and 5?

	individual 2	individual 4
A	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
B	The protein has an additional amino acid.	An amino acid in the protein has been deleted.
C	The protein has an additional amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
D	The protein has an additional positively charged amino acid.	An amino acid in the protein has been deleted.

29. Which of the following statement about stem cells is correct?

- A** Stem cells are specialised cells that have yet to express the genes and proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
- B** Stem cells are fully differentiated cells that reside under the surface of epithelia, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- C** Stem cells are undifferentiated embryonic cells that persist in the adult, and can give rise to all the cell types in the body.
- D** Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter that remains a stem cell, and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.

30. One type of genetically modified corn has

- A gene for the production of Bt toxin which protects the plant against a specific insect;
- A 'pat'-gene for tolerance to the herbicide 'Bastaa'. This gene is used to select plants with Bt toxin gene;
- An 'amp'-gene which was introduced in the plant together with the Bt toxin gene. This gene gives resistance to the antibiotic ampicillin.

There is concern that the 'amp' gene may transfer to enterobacteria in the human intestine during nucleic acid digestion making treatment with ampicillin ineffective for diseases caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

- A** An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.
- B** Bacteria took up DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.
- C** 50% of the enterobacteria isolated from humans are 'amp' resistant.
- D** All plant DNA is digested and destroyed in the human intestine during digestion of plant cells by enzymes including human nuclease enzymes.

ANSWERS

1	D	11	D	21	C
2	D	12	B	22	D
3	B	13	D	23	B
4	C	14	C	24	C
5	B	15	B	25	C
6	D	16	A	26	A
7	A	17	A	27	C
8	A	18	A	28	B
9	C	19	D	29	D
10	A	20	D	30	D